

Week 6: Representation – Gaussian Networks

Overview, Resources and Notes

1 Overview

So far in this module, we have exclusively looked at PGMs with *discrete* random variables (RVs). We started out by defining factors over discrete RVs (as discrete tables), as well as defining five factor operations: marginalise, multiply, divide, observe and reduce, and normalise. We then used the factors in Bayes networks (BNs) and Markov networks (MNs) to construct probabilistic models, and we used the factors and factor operations in cluster graphs (CGs) to perform inference (either the belief-propagation (BP) or the max-product algorithm).

We now move on to PGMs with *continuous* RVs, and more specifically Gaussian RVs with linear relationships between them. In this week we repeat the topics of the first term but we apply them to Gaussian networks: defining Gaussian factors, defining the five factor operations for Gaussian factors, constructing Gaussian BNs and MNs, and performing belief propagation (BP) in Gaussian networks. Luckily, we can reuse many of the concepts and techniques of the first term – we basically only have to define Gaussian factors and factor operations, and we can then reuse our previous approach to modelling and inference almost unchanged.

2 Resources

For this week, the prescribed resources again do not paint a very coherent picture, which might confuse you. For this reason, I recommend that you first study the notes (Section 3) and the example (Section 4), and then use the prescribed resources to fill in the gaps and to gain a wider perspective.

Prescribed resources

Study the following resources before the lectures and practical on Thursday (see the notes further down to get an idea of what to focus on):

- **Bishop** Section 8.1.4
- **Murphy** Sections 2.3, 4.2.3 and 4.3.5.1

Supplementary resources (optional)

I also recommend that you study the following resources, if you have access to it:

- **Koller&Friedman** Chapter 7 and Section 14.2

If you would like to read further, the following resources might be helpful: **Barber** Section 8.4; **Jordan** Chapter 13; **Murphy** Chapter 8.

Comments on the resources

- **Bishop** Section 8.1.4 describes linear-Gaussian models, which is similar to the Gaussian BNs described in the notes in Section 3. **Bishop**'s formulation provides an interesting perspective of these models, but we will rather use the formulation described in the notes in Section 3.

- **Murphy** Sections 2.3, 4.2.3 and 4.3.5 cover much of the same ground as the notes in Section 3, but contain useful visualisations, discussions, and alternative formulations. Specifically, note the following:
 - Section 2.3.1.1: Make sure that you understand the formulation and interpretation of the multivariate Gaussian distribution – you should already be familiar with it.
 - Section 2.3.1.2: The concept of a Gaussian shell is useful if you work with high-dimensional Gaussian distributions. However, we do not use it in this module.
 - Section 2.3.1.3: Make sure you can marginalise a Gaussian distribution in the mean-covariance form. You can ignore conditioning in the mean-covariance form – we will use the canonical form for it.
 - Section 2.3.1.4 introduces the canonical form, although with different notation than what we use; you can focus on the formulation in the notes in Section 3.
 - Sections 2.3.1.5 and 2.3.1.6: Although the matrix manipulations in these sections are useful to know, you do not have to know these proofs.
 - Section 2.3.2 discusses linear Gaussian systems, which is similar to our formulation in the notes in Section 3 (but with different notation). If you would like some practice in manipulating Gaussian distributions, you could study Sections 2.3.2.1 and 2.3.2.2, but you can otherwise ignore it. The example in Section 2.3.2.3 might give you some intuitive understanding – I can recommend it.
 - Section 2.3.3: This section is similar to our approach. Although it covers the same ground as the notes in Section 3, it is well worth studying.
 - Section 4.2.3 describes the same Gaussian BNs with linear relationships between RVs as given in the notes in Section 3, but formulated differently and using different notation. It also discusses the construction of the joint distribution for the whole model, which is something we tend to avoid. Take note of this section, but no need to study it.
 - Section 4.3.5.1: This section discusses *pairwise* Gaussian MNs, whereas our approach in the notes in Section 3 defines Gaussian factors over the *maximal cliques* of the MN. With a bit of work, one can see that the two approaches are equivalent, but we will stick to our formulation. However, what is important in this section is the link between conditional independence between RVs in the model and the structure of the information matrix of the joint distribution of the model. In sparsely connected graphs, the information matrix is sparse, which can be exploited in a number of ways. Also note that this sparsity is not present in the covariance matrix of the joint distribution (i.e., inverting a sparse matrix does not generally preserve the sparsity).

3 Notes

This section provides a coherent view of the approach that we take in this module to modelling and inference for Gaussian networks. Most of the material in this section is based on Chapter 7 and Section 14.2 of **Koller&Friedman**.

The topics covered in this section include the following:

- We first define multivariate Gaussian factors. We use two variants – the *mean-covariance* form and the *canonical* form, and we also discuss the conversion between them.
- We then define the five factor operations for Gaussian factors.
- Given the definition of Gaussian factors, we discuss how to construct Gaussian networks (i.e., probabilistic models with Gaussian RVs, and specifically with linear relationships between the RVs). We look at both Markov networks (MNs) and Bayes networks (BNs), and for the latter we spend some time on modelling conditional distributions as Gaussian factors.
- Lastly, we discuss how to perform marginal inference (specifically belief propagation) in Gaussian networks.

Gaussian factors

Suppose $\mathbf{x} = [x_1, \dots, x_n]^T$ is a vector of n continuous RVs. We want to define a Gaussian factor over these RVs, $\phi(\mathbf{x}) = \phi(x_1, \dots, x_n)$. We use two representations: the commonly used *mean-covariance* (or *moment*) form, and the less common *canonical* (or *information*) form.

The mean-covariance form is the most common way in which a multivariate Gaussian distribution is defined. It is given by

$$\mathcal{N}(\mathbf{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{\frac{n}{2}} |\boldsymbol{\Sigma}|^{\frac{1}{2}}} \exp \left\{ -\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^T \boldsymbol{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu}) \right\}, \quad (1)$$

where $\boldsymbol{\mu}$ is the mean vector and $\boldsymbol{\Sigma}$ is the covariance matrix. The covariance matrix is symmetric, which means that $\boldsymbol{\Sigma}^T = \boldsymbol{\Sigma}$, and it is also positive definite, which means that $\mathbf{x}^T \boldsymbol{\Sigma} \mathbf{x} > 0$ for *any* vector $\mathbf{x}$. Note that mean-covariance form as given in Equation 1 is normalised (i.e., it integrates to one), which means that it represents a valid probability distribution. The mean-covariance form is relatively easy to interpret: the maximum likelihood of the distribution is located at the mean, and the covariance matrix describes the shape of the distribution around the mean¹.

A less common form, but one that makes some of the factor operations much easier to do, is the canonical form. Because of the relative ease with which we can perform factor operations on the canonical form, we mostly use this form during inference, and switch to the mean-covariance form when we want to interpret the results. The canonical form is given by

$$\mathcal{C}(\mathbf{x}; \mathbf{K}, \mathbf{h}, g) = \exp \left\{ -\frac{1}{2} \mathbf{x}^T \mathbf{K} \mathbf{x} + \mathbf{h}^T \mathbf{x} + g \right\}, \quad (2)$$

where $\mathbf{K}$ is the *information* (or *precision*) *matrix*, $\mathbf{h}$ is the *information vector* (or *precision-weighted mean*), and g is a constant. Just as for the covariance matrix $\boldsymbol{\Sigma}$, the information matrix $\mathbf{K}$ is symmetric, but differently from the covariance matrix, the information matrix is only restricted to be positive *semi*-definite (i.e., $\mathbf{x}^T \mathbf{K} \mathbf{x} \geq 0$ for any vector $\mathbf{x}$), which means that it is not necessarily invertible. Also note that the constant g does not have to be chosen such that the factor is normalised; in fact, if $\mathbf{K}$ is not invertible, then it is not possible to normalise the factor. The canonical form is therefore more general (or expressive) than the mean-covariance form.

We can convert the mean-covariance form to the canonical form. To see how to do this, let us rewrite the mean-covariance form of Equation 1 by multiplying out the quadratic expression in the exponent and moving the factor outside of the exponential function into the exponential function, which produces²

$$\begin{aligned} \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}, \boldsymbol{\Sigma}) &= \exp \left\{ -\frac{1}{2} \mathbf{x}^T \boldsymbol{\Sigma}^{-1} \mathbf{x} + \frac{1}{2} \mathbf{x}^T \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu} + \frac{1}{2} \boldsymbol{\mu}^T \boldsymbol{\Sigma}^{-1} \mathbf{x} - \frac{1}{2} \boldsymbol{\mu}^T \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu} + \log \left(\frac{1}{(2\pi)^{\frac{n}{2}} |\boldsymbol{\Sigma}|^{\frac{1}{2}}} \right) \right\} \\ &= \exp \left\{ -\frac{1}{2} \mathbf{x}^T \boldsymbol{\Sigma}^{-1} \mathbf{x} + \boldsymbol{\mu}^T \boldsymbol{\Sigma}^{-1} \mathbf{x} - \frac{1}{2} \boldsymbol{\mu}^T \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu} - \log \left((2\pi)^{\frac{n}{2}} |\boldsymbol{\Sigma}|^{\frac{1}{2}} \right) \right\}. \end{aligned} \quad (3)$$

When we compare this with the definition of the canonical form in Equation 2, we can identify the information matrix as

$$\mathbf{K} = \boldsymbol{\Sigma}^{-1}, \quad (4)$$

the information vector as

$$\mathbf{h} = (\boldsymbol{\mu}^T \boldsymbol{\Sigma}^{-1})^T = (\boldsymbol{\Sigma}^{-1})^T \boldsymbol{\mu} = \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}, \quad (5)$$

and the constant as

$$g = -\frac{1}{2} \boldsymbol{\mu}^T \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu} - \log \left((2\pi)^{\frac{n}{2}} |\boldsymbol{\Sigma}|^{\frac{1}{2}} \right). \quad (6)$$

If the canonical form is normalised, then we can convert it to the mean-covariance form. We do this by first determining the covariance matrix from Equation 4,

$$\boldsymbol{\Sigma} = (\boldsymbol{\Sigma}^{-1})^{-1} = \mathbf{K}^{-1}, \quad (7)$$

and then using this in combination with Equation 5 to write

$$\boldsymbol{\Sigma}^{-1} \boldsymbol{\mu} = \mathbf{h} \Rightarrow \boldsymbol{\Sigma} \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu} = \mathbf{I} \boldsymbol{\mu} = \boldsymbol{\mu} = \boldsymbol{\Sigma} \mathbf{h} \Rightarrow \boldsymbol{\mu} = \mathbf{K}^{-1} \mathbf{h}. \quad (8)$$

Note that it is always possible to convert the mean-covariance form to the canonical form, but if the information matrix $\mathbf{K}$ is not invertible, then it is not possible to convert the canonical form to the mean-covariance form.

You should be familiar with the mean-covariance and canonical forms, as well as the conversions between them, but you do not need to memorise the equations; the definition of the different forms and the conversion between them will be given on a formula sheet attached to tests.

¹From previous modules, you should be able to interpret the diagonal and off-diagonal entries of the covariance matrix, know how to calculate the correlation between RVs from the covariance matrix, understand the relationship between correlation and independence for Gaussian RVs, and be able to visualise a 2-D Gaussian distribution given its mean vector and covariance matrix. If you cannot do these things, revise the relevant material from previous modules.

²Note that we use the fact here that $\mathbf{x}^T \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu} = (\mathbf{x}^T \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu})^T = \boldsymbol{\mu}^T (\boldsymbol{\Sigma}^{-1})^T \mathbf{x} = \boldsymbol{\mu}^T \boldsymbol{\Sigma}^{-1} \mathbf{x}$.

Gaussian factor operations

The standard factor operations – multiply, divide, marginalise³, observe and reduce, and normalise – are defined for Gaussian factors, and Gaussian factors are closed under these operations (i.e., when applied to Gaussian factors, these operations always produce Gaussian factors⁴). In this section we define the factor operations, but we do not derive them (their derivations are straightforward, but somewhat involved).

For factor operations that involve two factors, both the scopes and the order of the RVs in the Gaussian factors should correspond. Some of the other factor operations also require the vector of RVs to have a specific order. So, before we discuss the five factor operations, we first look at ways to change the scope and order of RVs for a factor.

If two factors involved in a factor operation do not have matching scopes, then the scopes can be matched by expanding the scope of one or both of the factors appropriately. To see how to do this, consider the following manipulation of a quadratic expression (found in the exponent of the exponential function of a Gaussian factor) in terms of the vector of RVs $\mathbf{x}$ and the RV y :

$$\begin{aligned}
 & -\frac{1}{2} \begin{bmatrix} \mathbf{x}^T & y \end{bmatrix} \begin{bmatrix} \mathbf{K} & \mathbf{0} \\ \mathbf{0}^T & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ y \end{bmatrix} + \begin{bmatrix} \mathbf{h}^T & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ y \end{bmatrix} \\
 &= -\frac{1}{2} \begin{bmatrix} \mathbf{x}^T & y \end{bmatrix} \begin{bmatrix} \mathbf{K}\mathbf{x} + \mathbf{0} \cdot y \\ \mathbf{0}^T\mathbf{x} + 0 \cdot y \end{bmatrix} + \mathbf{h}^T\mathbf{x} + 0 \cdot y \\
 &= -\frac{1}{2}\mathbf{x}^T\mathbf{K}\mathbf{x} - \frac{1}{2}\mathbf{x}^T\mathbf{0} \cdot y - \frac{1}{2}y \cdot \mathbf{0}^T\mathbf{x} - \frac{1}{2}y \cdot 0 \cdot y + \mathbf{h}^T\mathbf{x} + 0 \cdot y \\
 &= -\frac{1}{2}\mathbf{x}^T\mathbf{K}\mathbf{x} + \mathbf{h}^T\mathbf{x}.
 \end{aligned} \tag{9}$$

Note that the arrangements of zeros in the matrix and one of the vectors cause the expression to be unaffected by RV y . We can now use the manipulation of the quadratic expression of Equation 9 to expand the canonical factor in terms of random vector $\mathbf{x}$ to include the RV y as follows:

$$\begin{aligned}
 \mathcal{C}(\mathbf{x}; \mathbf{K}, \mathbf{h}, g) &= \exp \left\{ -\frac{1}{2}\mathbf{x}^T\mathbf{K}\mathbf{x} + \mathbf{h}^T\mathbf{x} + g \right\} \\
 &= \exp \left\{ -\frac{1}{2} \begin{bmatrix} \mathbf{x}^T & y \end{bmatrix} \begin{bmatrix} \mathbf{K} & \mathbf{0} \\ \mathbf{0}^T & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ y \end{bmatrix} + \begin{bmatrix} \mathbf{h}^T & 0 \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ y \end{bmatrix} + g \right\} \\
 &= \mathcal{C} \left(\begin{bmatrix} \mathbf{x} \\ y \end{bmatrix}; \begin{bmatrix} \mathbf{K} & \mathbf{0} \\ \mathbf{0}^T & 0 \end{bmatrix}, \begin{bmatrix} \mathbf{h} \\ 0 \end{bmatrix}, g \right).
 \end{aligned} \tag{10}$$

This means that in order to add an RV to the scope of a canonical factor (in a way that leaves the factor unaffected), one can simply pad the information matrix and information vector with zeros.

Some of the factor operations for Gaussian factors require the vector of RVs to have a specific order. To achieve a specific ordering of RVs, we can simply repeatedly swap pairs of RVs. To see how to swap two RVs, consider the following canonical form in terms of RVs x , y and z , where RVs x and z are swapped:

$$\mathcal{C} \left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}; \begin{bmatrix} k_{xx} & k_{xy} & k_{xz} \\ k_{yx} & k_{yy} & k_{yz} \\ k_{zx} & k_{zy} & k_{zz} \end{bmatrix}, \begin{bmatrix} h_x \\ h_y \\ h_z \end{bmatrix}, g \right) = \mathcal{C} \left(\begin{bmatrix} z \\ y \\ x \end{bmatrix}; \begin{bmatrix} k_{zz} & k_{zy} & k_{zx} \\ k_{yz} & k_{yy} & k_{yx} \\ k_{xz} & k_{xy} & k_{xx} \end{bmatrix}, \begin{bmatrix} h_z \\ h_y \\ h_x \end{bmatrix}, g \right). \tag{11}$$

Note that for the information matrix, the first and third rows are swapped *and* the first and third columns are swapped. For the information vector, the first and third elements are swapped. It should be obvious how to apply this idea to reorder RVs for factors with different numbers and orderings of RVs.

Let us now look at how the five factor operations are defined for Gaussian factors. Although the factor operations can be defined for the mean-covariance form, we only focus on the canonical form.

Multiplication: When two canonical forms (with RVs matched in both scope and order) are multiplied, the result is a canonical form where the parameters are summed, or

$$\mathcal{C}(\mathbf{x}; \mathbf{K}_1, \mathbf{h}_1, g_1) \cdot \mathcal{C}(\mathbf{x}; \mathbf{K}_2, \mathbf{h}_2, g_2) = \mathcal{C}(\mathbf{x}; \mathbf{K}_1 + \mathbf{K}_2, \mathbf{h}_1 + \mathbf{h}_2, g_1 + g_2). \tag{12}$$

³For continuous random variables, marginalisation involves integration, not summation as for discrete random variables.

⁴The only exception is the division operation, which could result in a information matrix $\mathbf{K}$ that is not positive semidefinite. However, under normal inference conditions, this issue should not arise.

Division: When two canonical forms (with matching RV scope and order) are divided, the result is a canonical form where the parameters of the denominator are subtracted from that of the numerator, or

$$\frac{\mathcal{C}(\mathbf{x}; \mathbf{K}_1, \mathbf{h}_1, g_1)}{\mathcal{C}(\mathbf{x}; \mathbf{K}_2, \mathbf{h}_2, g_2)} = \mathcal{C}(\mathbf{x}; \mathbf{K}_1 - \mathbf{K}_2, \mathbf{h}_1 - \mathbf{h}_2, g_1 - g_2). \quad (13)$$

Marginalisation: Consider the following canonical form, where the vector of RVs is partitioned into a vector $\mathbf{x}$ and a vector $\mathbf{y}$:

$$\mathcal{C}\left(\begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix}; \mathbf{K}, \mathbf{h}, g\right). \quad (14)$$

The information matrix is partitioned according to $\mathbf{x}$ and $\mathbf{y}$ as follows:

$$\mathbf{K} = \begin{bmatrix} \mathbf{K}_{xx} & \mathbf{K}_{xy} \\ \mathbf{K}_{yx} & \mathbf{K}_{yy} \end{bmatrix}, \quad (15)$$

and the information vector is also partitioned according to $\mathbf{x}$ and $\mathbf{y}$ as follows:

$$\mathbf{h} = \begin{bmatrix} \mathbf{h}_x \\ \mathbf{h}_y \end{bmatrix}. \quad (16)$$

Marginalisation for continuous RVs amounts to integration. If the RVs in the vector $\mathbf{y}$ are marginalised out, then it produces a canonical form

$$\int \mathcal{C}\left(\begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix}; \mathbf{K}, \mathbf{h}, g\right) d\mathbf{y} = \mathcal{C}(\mathbf{x}; \mathbf{K}', \mathbf{h}', g'), \quad (17)$$

where the parameters of the resultant canonical form are given by

$$\begin{aligned} \mathbf{K}' &= \mathbf{K}_{xx} - \mathbf{K}_{xy}\mathbf{K}_{yy}^{-1}\mathbf{K}_{yx} \\ \mathbf{h}' &= \mathbf{h}_x - \mathbf{K}_{xy}\mathbf{K}_{yy}^{-1}\mathbf{h}_y \\ g' &= g + \frac{1}{2}(\log |2\pi\mathbf{K}_{yy}^{-1}| + \mathbf{h}_y^T\mathbf{K}_{yy}^{-1}\mathbf{h}_y). \end{aligned} \quad (18)$$

Observe and reduce: This operation is somewhat similar to marginalisation. Consider the same partitioned canonical form described in Equations 14 to 16. If the RVs in the vector $\mathbf{y}$ are observed as $\hat{\mathbf{y}}$, then it produces a canonical form

$$\mathcal{C}\left(\begin{bmatrix} \mathbf{x} \\ \mathbf{y} = \hat{\mathbf{y}} \end{bmatrix}; \mathbf{K}, \mathbf{h}, g\right) = \mathcal{C}(\mathbf{x}; \mathbf{K}', \mathbf{h}', g'), \quad (19)$$

where the parameters in the resultant canonical form are given by

$$\begin{aligned} \mathbf{K}' &= \mathbf{K}_{xx} \\ \mathbf{h}' &= \mathbf{h}_x - \mathbf{K}_{xy}\hat{\mathbf{y}} \\ g' &= g + \mathbf{h}_y^T\hat{\mathbf{y}} - \frac{1}{2}\hat{\mathbf{y}}^T\mathbf{K}_{yy}\hat{\mathbf{y}}. \end{aligned} \quad (20)$$

Normalisation: To normalise a canonical form, we use the fact that the constant calculated in Equation 6 corresponds to a normalised distribution. For the canonical form to be normalised, we write this desired constant in terms of the information matrix $\mathbf{K}$ and information vector $\mathbf{h}$, using Equations 7 and 8 as follows:

$$\begin{aligned} g &= -\frac{1}{2}\boldsymbol{\mu}^T\boldsymbol{\Sigma}^{-1}\boldsymbol{\mu} - \log\left((2\pi)^{\frac{n}{2}}|\boldsymbol{\Sigma}|^{\frac{1}{2}}\right) \\ &= -\frac{1}{2}(\mathbf{K}^{-1}\mathbf{h})^T(\mathbf{K}^{-1})^{-1}\mathbf{K}^{-1}\mathbf{h} - \log\left((2\pi)^{\frac{n}{2}}|\mathbf{K}^{-1}|^{\frac{1}{2}}\right) \\ &= -\frac{1}{2}\mathbf{h}^T\mathbf{K}^{-1}\mathbf{K}\mathbf{K}^{-1}\mathbf{h} - \log\left((2\pi)^{\frac{n}{2}}|\mathbf{K}|^{-\frac{1}{2}}\right) \\ &= -\frac{1}{2}\mathbf{h}^T\mathbf{K}^{-1}\mathbf{h} - \log\left((2\pi)^{\frac{n}{2}}|\mathbf{K}|^{-\frac{1}{2}}\right). \end{aligned} \quad (21)$$

To normalise a canonical form, we therefore simply replace the existing constant g with the desired one calculated in Equation 21.

You should be familiar with these operations, but you do not need to memorise the expressions involved. The first four factor operations are made available on the formula sheet during tests. We usually do not bother to do normalisation, since we are typically only interested in the “shape” of the distribution (i.e., we are only interested in the *relative* likelihoods at different locations). For this reason, we often do not bother to calculate the constant g , since it does not influence the shape of the distribution.

Gaussian Markov networks (MNs) and Bayes networks (BNs)

With the description of Gaussian factors and factor operations in hand, let us now look at how to construct a probabilistic model with Gaussian RVs. We first look at the easy case (MNs) and then at the slightly more complex case (BNs).

Markov networks (MNs): Constructing a Gaussian MN is pretty simple: After defining the RVs and specifying the graph (in the same way as for MNs with discrete RVs), we define each factor required by the graph (i.e., for each maximal clique) as a Gaussian factor in either the mean-covariance form (Equation 1) or the canonical form (Equation 2). That’s it. Easy.

Bayes networks (BNs): Constructing a Gaussian BN is a bit more involved, mainly because we need to represent conditional Gaussian distributions as Gaussian factors. To be specific, let’s consider the structure $\mathbf{x} \rightarrow \mathbf{y}$; that is, a node containing the random vector $\mathbf{y}$ with the node containing the random vector $\mathbf{x}$ as its parent.⁵ We assume that there are k elements in random vector $\mathbf{y}$ (this is important later). Except for the definition of the marginal distribution $p(\mathbf{x})$ (which can easily be represented as a Gaussian factor in either the mean-covariance or canonical form), this structure also requires the definition of the conditional distribution $p(\mathbf{y}|\mathbf{x})$.⁶ The challenge is to represent this conditional distribution as a Gaussian factor in either the mean-covariance or the canonical form.

In this module, we only look at linear relationships between RVs and their parents in the graph – these relationships are called *linear Gaussians*. Such a relationship between random vector $\mathbf{y}$ and its parent $\mathbf{x}$ is given by

$$\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{b} + \mathbf{n}, \quad (22)$$

where $\mathbf{A}$ is a constant matrix, $\mathbf{b}$ is a constant vector, and $\mathbf{n}$ is a vector of zero-mean Gaussian noise, or

$$\mathbf{n} \sim \mathcal{N}(\mathbf{n}; \mathbf{0}, \Sigma_{\mathbf{n}}). \quad (23)$$

We also assume that the noise $\mathbf{n}$ is independent of the random vector $\mathbf{x}$. Note that although $\mathbf{n}$ is a vector of RVs, we do not include it as a node in the graph – it is simply used to describe the relationship between random vectors $\mathbf{x}$ and $\mathbf{y}$, and we can think of it as a vector of “auxiliary” RVs⁷

Let us now look at how to represent this linear Gaussian as the conditional distribution $p(\mathbf{y}|\mathbf{x})$. It can be shown that a linear transformation of Gaussian RVs (such as Equation 22) produces a Gaussian distribution. We now need to find the parameters (mean and covariance) of this Gaussian distribution. The mean of this conditional distribution over $\mathbf{y}$ is calculated as

$$\mathcal{E}[\mathbf{y}|\mathbf{x}] = \mathcal{E}[\mathbf{A}\mathbf{x} + \mathbf{b} + \mathbf{n}|\mathbf{x}] = \mathbf{A}\mathbf{x} + \mathbf{b} + \mathcal{E}[\mathbf{n}] = \mathbf{A}\mathbf{x} + \mathbf{b} \quad (24)$$

and the covariance of is calculated as

$$\begin{aligned} \text{Cov}[\mathbf{y}|\mathbf{x}] &= \mathcal{E}[(\mathbf{y} - \mathcal{E}[\mathbf{y}|\mathbf{x}]) (\mathbf{y} - \mathcal{E}[\mathbf{y}|\mathbf{x}])^T | \mathbf{x}] \\ &= \mathcal{E}[(\mathbf{A}\mathbf{x} + \mathbf{b} + \mathbf{n} - (\mathbf{A}\mathbf{x} + \mathbf{b})) (\mathbf{A}\mathbf{x} + \mathbf{b} + \mathbf{n} - (\mathbf{A}\mathbf{x} + \mathbf{b}))^T | \mathbf{x}] \\ &= \mathcal{E}[\mathbf{n}\mathbf{n}^T] \\ &= \Sigma_{\mathbf{n}}. \end{aligned} \quad (25)$$

We can therefore write the conditional distribution of $\mathbf{y}$ given $\mathbf{x}$ as the Gaussian distribution

$$p(\mathbf{y}|\mathbf{x}) = \mathcal{N}(\mathbf{y}; \mathbf{A}\mathbf{x} + \mathbf{b}, \Sigma_{\mathbf{n}}). \quad (26)$$

It might seem that we have reached our goal of describing the conditional distribution $p(\mathbf{y}|\mathbf{x})$ as a Gaussian factor. However, note that the scope of the factor is $[\mathbf{x} \ \mathbf{y}]^T$, and the conditional distribution in Equation 26 is not given in the mean-covariance or canonical form over this (full) scope. So we need to do a bit more work.

⁵Note that this is the first time that we lump several RVs together in a node of a BN. This can be shown to be equivalent to a graph structure where each $y_i \in \mathbf{y}$ is located in its own node, and it has *all* the RVs contained in the random vector $\mathbf{x}$ as its parents. The structure $\mathbf{x} \rightarrow \mathbf{y}$ is therefore just a “summary” of the relationship between several RVs, which is especially convenient in Gaussian networks.

⁶Note that although this representation seems to be specific to a node with only one parent, it can also be used for nodes with multiple parents – the random vector $\mathbf{x}$ just has to include *all* the parent RVs of the random vector $\mathbf{y}$.

⁷Formally, we define $\mathbf{n}$ as RVs of the model, but we marginalise them out in the process of the constructing the factor over $\mathbf{x}$ and $\mathbf{y}$.

It turns out that such a linear Gaussian cannot be represented in the mean-covariance form (the information matrix $\mathbf{K}$ for this factor is not invertible), and so we work towards the canonical form. The first step is to use the definition of the multivariate Gaussian distribution (Equation 1) to write $p(\mathbf{y}|\mathbf{x})$ in terms of the exponential function, using manipulations similar to Equation 3 to write Equation 26 as⁸:

$$\begin{aligned} p(\mathbf{y}|\mathbf{x}) &= \frac{1}{(2\pi)^{\frac{k}{2}} |\boldsymbol{\Sigma}_{\mathbf{n}}|^{\frac{1}{2}}} \exp \left\{ -\frac{1}{2} (\mathbf{y} - (\mathbf{A}\mathbf{x} + \mathbf{b}))^T \boldsymbol{\Sigma}_{\mathbf{n}}^{-1} (\mathbf{y} - (\mathbf{A}\mathbf{x} + \mathbf{b})) \right\} \\ &= \exp \left\{ -\frac{1}{2} (\mathbf{y} - \mathbf{A}\mathbf{x})^T \boldsymbol{\Sigma}_{\mathbf{n}}^{-1} (\mathbf{y} - \mathbf{A}\mathbf{x}) + \mathbf{b}^T \boldsymbol{\Sigma}_{\mathbf{n}}^{-1} (\mathbf{y} - \mathbf{A}\mathbf{x}) - \frac{1}{2} \mathbf{b}^T \boldsymbol{\Sigma}_{\mathbf{n}}^{-1} \mathbf{b} - \log \left((2\pi)^{\frac{k}{2}} |\boldsymbol{\Sigma}_{\mathbf{n}}|^{\frac{1}{2}} \right) \right\}. \end{aligned} \quad (27)$$

We then use the fact

$$\mathbf{y} - \mathbf{A}\mathbf{x} = \begin{bmatrix} -\mathbf{A} & \mathbf{I} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} \quad (28)$$

and use this to rewrite the first term in the exponent of Equation 27 as

$$\begin{aligned} -\frac{1}{2} (\mathbf{y} - \mathbf{A}\mathbf{x})^T \boldsymbol{\Sigma}_{\mathbf{n}}^{-1} (\mathbf{y} - \mathbf{A}\mathbf{x}) &= -\frac{1}{2} \begin{bmatrix} \mathbf{x}^T & \mathbf{y}^T \end{bmatrix} \begin{bmatrix} -\mathbf{A}^T \\ \mathbf{I} \end{bmatrix} \boldsymbol{\Sigma}_{\mathbf{n}}^{-1} \begin{bmatrix} -\mathbf{A} & \mathbf{I} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} \\ &= -\frac{1}{2} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix}^T \begin{bmatrix} \mathbf{A}^T \boldsymbol{\Sigma}_{\mathbf{n}}^{-1} \mathbf{A} & -\mathbf{A}^T \boldsymbol{\Sigma}_{\mathbf{n}}^{-1} \\ -\boldsymbol{\Sigma}_{\mathbf{n}}^{-1} \mathbf{A} & \boldsymbol{\Sigma}_{\mathbf{n}}^{-1} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix}. \end{aligned} \quad (29)$$

We also rewrite the second term in the exponent of Equation 27 as

$$\mathbf{b}^T \boldsymbol{\Sigma}_{\mathbf{n}}^{-1} (\mathbf{y} - \mathbf{A}\mathbf{x}) = \mathbf{b}^T \boldsymbol{\Sigma}_{\mathbf{n}}^{-1} \begin{bmatrix} -\mathbf{A} & \mathbf{I} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} = \left(\begin{bmatrix} -\mathbf{A}^T \boldsymbol{\Sigma}_{\mathbf{n}}^{-1} \\ \boldsymbol{\Sigma}_{\mathbf{n}}^{-1} \end{bmatrix} \mathbf{b} \right)^T \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix}. \quad (30)$$

At this point, the exponent of Equation 27 is written as a quadratic expression in terms of the vector $\begin{bmatrix} \mathbf{x} & \mathbf{y} \end{bmatrix}^T$, which is the format used by the definition of the canonical form in Equation 2. We can therefore write $p(\mathbf{y}|\mathbf{x})$ as the canonical form

$$p(\mathbf{y}|\mathbf{x}) = \mathcal{C} \left(\begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix}; \mathbf{K}_{\mathbf{y}|\mathbf{x}}, \mathbf{h}_{\mathbf{y}|\mathbf{x}}, g_{\mathbf{y}|\mathbf{x}} \right) \quad (31)$$

where its parameters can be read from Equations 27, 29 and 30, and are given by

$$\begin{aligned} \mathbf{K}_{\mathbf{y}|\mathbf{x}} &= \begin{bmatrix} \mathbf{A}^T \boldsymbol{\Sigma}_{\mathbf{n}}^{-1} \mathbf{A} & -\mathbf{A}^T \boldsymbol{\Sigma}_{\mathbf{n}}^{-1} \\ -\boldsymbol{\Sigma}_{\mathbf{n}}^{-1} \mathbf{A} & \boldsymbol{\Sigma}_{\mathbf{n}}^{-1} \end{bmatrix} \\ \mathbf{h}_{\mathbf{y}|\mathbf{x}} &= \begin{bmatrix} -\mathbf{A}^T \boldsymbol{\Sigma}_{\mathbf{n}}^{-1} \\ \boldsymbol{\Sigma}_{\mathbf{n}}^{-1} \end{bmatrix} \mathbf{b} \\ g_{\mathbf{y}|\mathbf{x}} &= -\frac{1}{2} \mathbf{b}^T \boldsymbol{\Sigma}_{\mathbf{n}}^{-1} \mathbf{b} - \log \left((2\pi)^{\frac{k}{2}} |\boldsymbol{\Sigma}_{\mathbf{n}}|^{\frac{1}{2}} \right). \end{aligned} \quad (32)$$

We have therefore succeeded in writing the linear Gaussian $p(\mathbf{y}|\mathbf{x})$ as a Gaussian factor. For hand calculations, this representation works well; however, since $\mathbf{K}_{\mathbf{y}|\mathbf{x}}$ is not invertible, it creates numerical issues if we want to implement it in `emdw`. We will explore ways to address these numerical issues in the practical assignment for this week.

In this section, we have only looked at representing the conditional distribution required by a single directed edge in a BN. However, it should be obvious that for a BN with multiple directed edges, all the conditional distributions required by the graph can be represented in the same way. For BNs where all the RVs are Gaussian and all the relationships between the RVs are linear, we therefore now have the ability to fully describe such a probabilistic model.

Note that I consider the ability to do the manipulations from Equation 22 to Equation 32 important, so you have to know it. However, the results of Equations 31 and 32 will be available on the formula sheet for tests.

⁸Remember that k is the number of elements in random vector $\mathbf{y}$.

Belief propagation in Gaussian networks

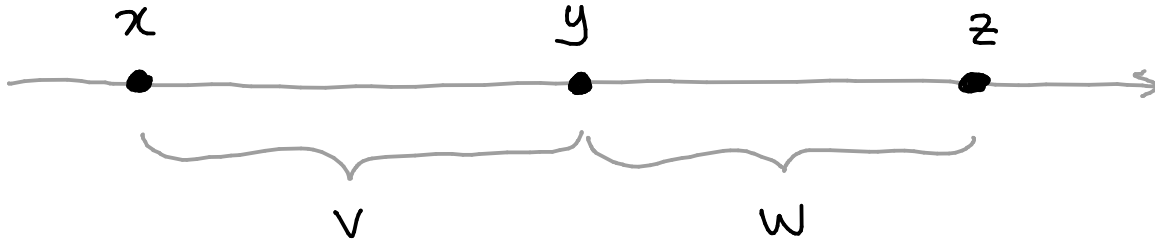
If we have a model described in terms of Gaussian factors, we can construct a cluster graph and apply (loopy) belief propagation as before (except that marginalisation is done via integration and not summation), and query the cluster graph once message passing has converged. In such a cluster graph, the cluster potentials, messages and beliefs are all represented by Gaussian factors.

In principle, such inference is simple to perform. However, if there are linear Gaussian factors in the model, we could encounter numerical issues in the implementation. In this week's practical, we will explore ways to address the numerical issues.

4 Example

Let's use some of these concepts by working through an example.

Consider the following setup: Suppose there are 3 points with their (1-D) locations denoted by x , y and z respectively. We have a fairly good idea of the locations x and z , but we do not know anything about the location y . We also have a noisy measurement of the difference between x and y – denoted by v – and a noisy measurement of the difference between y and z – denoted by w . We would like to infer the location y from the given information. The problem setup is depicted below.



Let's make the information available to us more precise. The knowledge that we have about location x is represented by a Gaussian distribution with mean 0 m and standard deviation 0.1 m, or

$$p(x) = \mathcal{N}(x; 0, 0.01). \quad (33)$$

Similarly, the knowledge about z is given by a Gaussian distribution with mean 1 m and standard deviation of 0.1 m, or

$$p(z) = \mathcal{N}(z; 1, 0.01). \quad (34)$$

The measurement v is modelled as

$$v = y - x + n_v, \quad (35)$$

where n_v is zero-mean noise with a standard deviation of 0.1 m, or

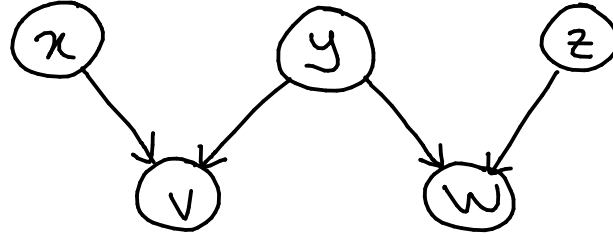
$$n_v \sim \mathcal{N}(n_v; 0, 0.01). \quad (36)$$

Note that n_v is considered to be an auxiliary random variable – it is used to describe the relationship between x , y , and v , but we are not interested in performing inference over it. The auxiliary random variables are used to set up the factors of the model, but they will not be retained further. Similarly to the measurement v , the measurement w is modelled as

$$w = z - y + n_w, \quad n_w \sim \mathcal{N}(n_w; 0, 0.01), \quad (37)$$

where n_w is also considered an auxiliary random variable. The values of the measurements are $v = 0.61$ m and $w = 0.41$ m. Since x is around 0 m, y is around 1 m, and y is measured to be around 60% of the way from x to z , we expect y to be about 0.6 m.

Let's now create a Bayes network model of this problem. Since locations x and y cause the measurement v , and locations y and z cause the measurement w , an obvious Bayes network structure is the following:



Note that the auxiliary random variables, n_v and n_w , are not included in the Bayes network. The factors required by this Bayes network are $\phi_1(x) = p(x)$, $\phi_2(y) = p(y)$, $\phi_3(z) = p(z)$, $\phi_4(x, y, v) = p(v|x, y)$ and $\phi_5(y, z, w) = p(w|y, z)$. Factor $\phi_1(x) = p(x)$ is defined in Equation 33; when we convert it to the canonical form⁹, we get¹⁰

$$\phi_1(x) = \mathcal{C}(x; 100, 0, -). \quad (38)$$

Similarly, factor $\phi_3(z) = p(z)$ is given by

$$\phi_3(z) = \mathcal{C}(z; 100, 100, -). \quad (39)$$

We have no knowledge of the location y , which can be represented by the vacuous Gaussian factor¹¹

$$\phi_2(y) = \mathcal{C}(y; 0, 0, 0). \quad (40)$$

For the factor $\phi_4(x, y, v) = p(v|x, y)$, one can combine Equations 35 and 36 to form the conditional distribution for v given x and y in the mean-covariance form¹²:

$$\phi_4(x, y, v) = p(v|x, y) = \mathcal{N}(v; y - x, 0.01). \quad (41)$$

We would like to have this factor in the canonical form in terms of x , y and v . To do this, we first go to the definition of the Gaussian distribution in terms of the exponential function; i.e.,

$$\begin{aligned} \phi_4(x, y, v) &= \frac{1}{\sqrt{2\pi \cdot 0.01}} \exp \left\{ -\frac{1}{2} \frac{(v - (y - x))^2}{0.01} \right\} \\ &= \exp \left\{ -\frac{1}{2} (x - y + v) 100 (x - y + v) - \log(\sqrt{2\pi \cdot 0.01}) \right\} \end{aligned} \quad (42)$$

Now we use the fact that

$$x - y + v = \begin{bmatrix} x & y & v \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ v \end{bmatrix} \quad (43)$$

to write

$$\begin{aligned} \phi_4(x, y, v) &= \exp \left\{ -\frac{1}{2} \begin{bmatrix} x & y & v \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} 100 \begin{bmatrix} 1 & -1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ v \end{bmatrix} - \log(\sqrt{2\pi \cdot 0.01}) \right\} \\ &= \exp \left\{ -\frac{1}{2} \begin{bmatrix} x & y & v \end{bmatrix} \begin{bmatrix} 100 & -100 & 100 \\ -100 & 100 & -100 \\ 100 & -100 & 100 \end{bmatrix} \begin{bmatrix} x \\ y \\ v \end{bmatrix} - \log(\sqrt{2\pi \cdot 0.01}) \right\}. \end{aligned} \quad (44)$$

By comparing this to the definition of the canonical form in Equation 2, we can write down the canonical form of factor $\phi_4(x, y, v)$ as

$$\phi_4(x, y, v) = \mathcal{C} \left(\begin{bmatrix} x \\ y \\ v \end{bmatrix}; \begin{bmatrix} 100 & -100 & 100 \\ -100 & 100 & -100 \\ 100 & -100 & 100 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, -\log(\sqrt{2\pi \cdot 0.01}) \right). \quad (45)$$

⁹See Equation 2.

¹⁰Since the constant g does not influence the information matrix $\mathbf{K}$ or the information vector $\mathbf{h}$ during any of the factor operations used during inference, and we can always at the end normalise the beliefs using only the information matrix and vector, we usually do not bother calculating the constant g during modelling or inference.

¹¹You can see why this factor is vacuous (i.e., it “carries no information”) by realising that it is equivalent to 1 – if you multiply it with any other factor, it will leave the other factor unchanged.

¹²Make sure you understand how to do this conversion – it is here that we “get rid of” the auxiliary random variable.

We could have arrived at the same result by rewriting Equation 35 as

$$v = \begin{bmatrix} -1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + n_v, \quad (46)$$

and comparing this and Equation 36 with the general description of the linear Gaussian of Equations 22 and 23. From this comparison, we could have identified the parameters of the linear Gaussian as $\mathbf{A} = \begin{bmatrix} -1 & 1 \end{bmatrix}$, $\mathbf{b} = 0$, $\Sigma_n = 0.01$, and $k = 1$. We could then have used the parameters for the linear Gaussian factor in canonical form given in Equation 32 to write factor $\phi_4(x, y, v)$ as

$$\phi_4(x, y, v) = \mathcal{C} \left(\begin{bmatrix} x \\ y \\ v \end{bmatrix}; \left[\begin{array}{c|c} \begin{bmatrix} -1 \\ 1 \end{bmatrix} \frac{1}{0.01} \begin{bmatrix} -1 & 1 \end{bmatrix} & -\begin{bmatrix} -1 \\ 1 \end{bmatrix} \frac{1}{0.01} \\ \hline -\frac{1}{0.01} \begin{bmatrix} -1 & 1 \end{bmatrix} & \frac{1}{0.01} \end{array} \right], \left[-\begin{bmatrix} -1 \\ 1 \end{bmatrix} \frac{1}{0.01} \right] \cdot 0, -\log \left((2\pi)^{\frac{1}{2}} 0.01^{\frac{1}{2}} \right) \right), \quad (47)$$

which reduces to the same factor as given in Equation 45.

Let's now insert the evidence $v = 0.61$ m into factor $\phi_4(x, y, v)$ ¹³:

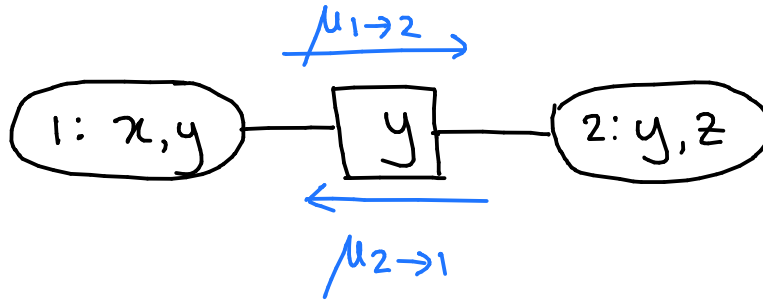
$$\begin{aligned} \phi_4(x, y) &= \mathcal{C} \left(\begin{bmatrix} x \\ y \end{bmatrix}; \begin{bmatrix} 100 & -100 \\ -100 & 100 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 100 \\ -100 \end{bmatrix} 0.61, - \right) \\ &= \mathcal{C} \left(\begin{bmatrix} x \\ y \end{bmatrix}; \begin{bmatrix} 100 & -100 \\ -100 & 100 \end{bmatrix}, \begin{bmatrix} -61 \\ 61 \end{bmatrix}, - \right). \end{aligned} \quad (48)$$

Similarly to factor $\phi_4(x, y)$, we can construct factor $\phi_5(y, z, w)$ in the canonical form and then insert the evidence $w = 0.41$ m; this results in¹⁴

$$\phi_5(y, z) = \mathcal{C} \left(\begin{bmatrix} y \\ z \end{bmatrix}; \begin{bmatrix} 100 & -100 \\ -100 & 100 \end{bmatrix}, \begin{bmatrix} -41 \\ 41 \end{bmatrix}, - \right). \quad (49)$$

All 5 factors are now described in the canonical form, and we are therefore done with the modelling step.

For inference, we define a cluster graph with two clusters – cluster 1 has scope $\{x, y\}$ and cluster 2 has scope $\{y, z\}$, and they are connected with an edge with sepset $\{y\}$; this cluster graph is shown below.



We assign factors $\phi_1(x)$ and $\phi_4(x, y)$ to cluster 1 and factors $\phi_3(z)$ and $\phi_5(y, z)$ to cluster 2¹⁵. The cluster potential of cluster 1 is therefore given by¹⁶

$$\psi_1(x, y) = \phi_1(x)\phi_4(x, y) = \mathcal{C} \left(\begin{bmatrix} x \\ y \end{bmatrix}; \begin{bmatrix} 200 & -100 \\ -100 & 100 \end{bmatrix}, \begin{bmatrix} -61 \\ 61 \end{bmatrix}, - \right) \quad (50)$$

and the cluster potential of cluster 2 is given by

$$\psi_2(y, z) = \phi_3(z)\phi_5(y, z) = \mathcal{C} \left(\begin{bmatrix} y \\ z \end{bmatrix}; \begin{bmatrix} 100 & -100 \\ -100 & 200 \end{bmatrix}, \begin{bmatrix} -41 \\ 141 \end{bmatrix}, - \right). \quad (51)$$

¹³See Equations 19 and 20.

¹⁴Make sure you can get the same result.

¹⁵Since $\phi_2(y)$ is vacuous, we can assign it to either cluster or omit it – any of these options will have the same effect.

¹⁶See Equation 12 for the multiplication operation.

Let's use belief propagation for inference. The message from cluster 1 to cluster 2 is given by¹⁷

$$\begin{aligned}\mu_{1 \rightarrow 2}(y) &= \int \psi_1(x, y) dx = \mathcal{C} \left(y; 100 - (-100) \frac{1}{200}(-100), 61 - (-100) \frac{1}{200}(-61), - \right) \\ &= \mathcal{C}(y; 50, 30.5, -).\end{aligned}\quad (52)$$

Similarly, the message from cluster 2 to cluster 1 is given by

$$\begin{aligned}\mu_{2 \rightarrow 1}(y) &= \int \psi_2(y, z) dz = \mathcal{C} \left(y; 100 - (-100) \frac{1}{200}(-100), -41 - (-100) \frac{1}{200}(141), - \right) \\ &= \mathcal{C}(y; 50, 29.5, -).\end{aligned}\quad (53)$$

The posterior belief over the location y is given by the sepset belief, which is calculated as¹⁸

$$\beta_{1,2}(y) = \mu_{1 \rightarrow 2}(y) \mu_{2 \rightarrow 1}(y) = \mathcal{C}(y; 50 + 50, 30.5 + 29.5, -) = \mathcal{C}(y; 100, 60, -).\quad (54)$$

To interpret this posterior belief over y , we can convert it to the mean-covariance form, which produces

$$\beta_{1,2}(y) = \mathcal{N}(y; 0.6, 0.01).\quad (55)$$

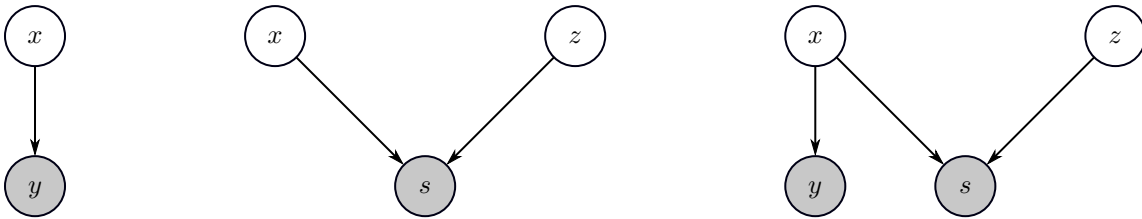
This result corresponds to our expectations about y given the information about the problem.

What you can hopefully see from this example is that the modelling and inference techniques we studied so far can be applied to Gaussian networks; only the parameterisation of the factors and the manipulations involved in the factor operations differ from the discrete networks we've used previously¹⁹.

5 Homework

If you have worked through the notes in Section 3 and the example in Section 4, the concepts involved in Gaussian networks might not quite have solidified. For this reason, I recommend that you work through the following homework problem by yourself. [The answers are given on the last page in blue.](#)

The question involves modelling and inference for a Bayes network (BN) with Gaussian random variables. The approach of this question is to first perform modelling and inference for two very simple BNs (left and middle in diagram below), and then combine them to perform modelling and inference for a slightly more complex BN (right).



For the first BN (left), y is noisy measurement of the latent RV x , described by

$$y = x + v,\quad (56)$$

where v is zero-mean Gaussian noise, $v \sim \mathcal{N}(v; 0, 1)$, that is added to the latent RV x to form the measurement. Note that v is an auxiliary RV that we just use to set up the factor representing the relationship between x and y , but after that it “disappears”²⁰ We also have some prior knowledge about x , which is represented by the prior distribution

$$p(x) = \mathcal{N}(x; 0, 4).\quad (57)$$

The measurement is observed as $y = 1$.

¹⁷See Equations 17 and 18 for the marginalisation operation.

¹⁸The normalisation to produce a valid probability distribution is implied here.

¹⁹There are a few more complexities when working with Gaussian networks though – we'll encounter some of them later on.

²⁰Formally, v is marginalised out in the process of setting up the factor.

For the second BN (middle), s is a noisy measurement of the sum of latent RVs x and z , or

$$s = x + z + w, \quad w \sim \mathcal{N}(w; 0, 0.5). \quad (58)$$

The prior distribution $p(x)$ is given by Equation 57, and the prior distribution $p(z)$ is given by

$$p(z) = \mathcal{N}(z; 4, 10). \quad (59)$$

The measurement is observed as $s = 5$.

The third BN (right) is simply the combination of the first two BNs.

Our approach to inference in this question is the simple, naive approach: We construct all the factors, multiply them together to form the joint distribution, and manipulate this joint distribution to query the model.

- (a) Construct the factors of the first (left) BN: Convert $p(x)$ to the canonical form, use Equation 56 to construct $p(y|x)$ in the canonical form, and insert the evidence $y = 1$ to obtain $p(y = 1|x)$. Now perform inference to obtain the posterior marginal belief $p(x|y = 1)$: Multiply $p(x)$ and $p(y = 1|x)$ form the joint distribution $p(x, y = 1)$, and then obtain the query $p(x|y = 1)$ in the mean-covariance form²¹. Compare $p(x)$ and $p(x|y = 1)$ in the mean-covariance form – can you see how the measurement has changed the belief about x ?
- (b) Let's calculate $p(x, z|s = 5)$ for the second (middle) BN: You've already calculated $p(x)$ in the canonical form; now convert $p(z)$ to the canonical form. Next, calculate $p(s|x, z)$ as a factor in the canonical form. Then, observe the evidence in this factor to form $p(s = 5|x, z)$. Lastly, multiply $p(x)$, $p(z)$ and $p(s = 5|x, z)$ to form the joint distribution $p(x, z, s = 5)$ (which is proportional to $p(x, z|s = 5)$). Keep this result in the canonical form.
- (c) Let's now calculate the query $p(x|y = 1, s = 5)$ for the combined BN (the one on the right): You've already calculated all the factors in the canonical form; now combine them to form $p(x, z|y = 1, s = 5) \propto p(x)p(z)p(y = 1|x)p(s = 5|x, z)$. Marginalise out z , convert the result to the mean-covariance form, and compare with the results obtained in (a).

²¹Note that $p(x|y = 1) \propto p(x, y = 1)$, so you just have to normalise $p(x, y = 1)$ to obtain $p(x|y = 1)$. However, we usually do not bother to calculate the constant parameter g (which determines the normalisation of the distribution), which means that you do not have to explicitly perform normalisation.

Answers:

(a) $p(x|y = 1) = \mathcal{N}(x; 0.8, 0.8)$

(b) $p(x, z|s = 5) = \mathcal{C}\left(\begin{bmatrix} x \\ z \end{bmatrix}; \begin{bmatrix} 2.25 & 2 \\ 2 & 2.1 \end{bmatrix}, \begin{bmatrix} 10 \\ 10.4 \end{bmatrix}, -\right)$

(c) $p(x|y = 1, s = 5) = \mathcal{N}(x; 0.8142, 0.7434)$